Module 3: Pretraining Data Pipeline

From Raw Documents to Verified Token Batches

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Lecture map

1. From documents to tokens
2. Packing documents into examples
3. Storage and correctness

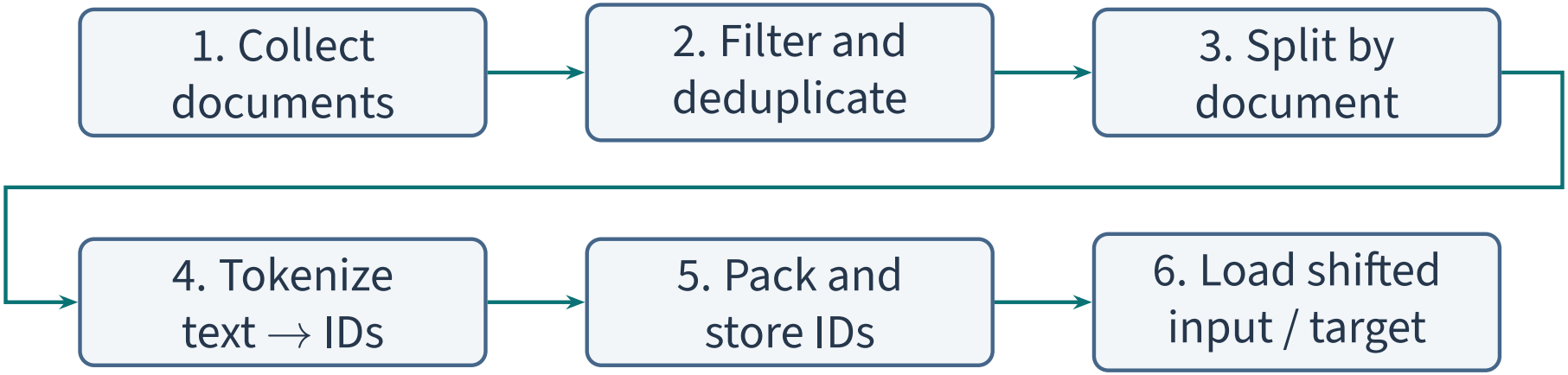
Guiding question

How do we turn raw documents into trustworthy next-token training batches?

What should be clear by the end?

1. Trace raw documents through filtering, tokenization, packing and batching.
2. Explain byte-level BPE and the vocabulary–sequence-length trade-off.
3. Pack documents while keeping boundaries explicit.
4. Verify binary storage, sampling, and the next-token shift.

The path from raw text to a training batch



Fit the tokenizer on the training split; process training and validation separately.

From yesterday’s baseline to today’s pipeline

Yesterday: byte baseline

UTF-8 bytes + EOT: 257 token IDs. The supplied runner packs separate train and validation shards.

Today: learned BPE

Fit merges on training documents; pack both splits with the same tokenizer; inspect boundaries and shifted batches.

What changes?

Token IDs, vocabulary size and sequence length change. Loss per byte-token and loss per BPE-token are not directly comparable.

PART 1

From documents to tokens

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Start with documents, not model inputs

A document

A piece of text plus its source and identity: a book, article, web page, or other record.

Extract usable text

Keep content; remove or inspect navigation, markup, duplicated page furniture, and extraction errors.

Where might web text come from?

Web archives store captured pages and related records (WARC); extracted-text files (WET) provide a text-oriented starting point. Neither is automatically clean.

Filter, deduplicate, then split by document

1. **Filter:** inspect rules for empty, short, or repetitive text; record what they remove.
2. **Deduplicate:** group repeated documents before splitting to reduce train–validation leakage.
3. **Split:** assign documents to train or validation, then keep their streams separate.

Fit only on training data

Learn tokenizer merges on the training partition; apply the resulting tokenizer to both partitions.

Granularity moves cost between vocabulary and sequence

Unit	Vocabulary	Sequence	Main cost
bytes	256	very long	attention and context
words	millions	short	embeddings and unknowns
subwords	tunable	moderate	learned merge system

Why byte-level BPE?

Every UTF-8 string decomposes into 256 byte values, so the base system has no unknown text. Frequent byte sequences can then become single tokens.

System consequence

Tokenization changes sequence length, attention work, KV-cache size, multilingual efficiency, and the number of embedding parameters.

BPE learns a greedy compression hierarchy

1. Start from byte IDs and reserved special tokens.
2. Count adjacent token pairs in the training corpus.
3. Merge the most frequent pair into one new token.
4. Repeat until the vocabulary reaches its target size.

$$(p_0, p_1) \mapsto v_{\text{new}}, \qquad \text{repr}(v_{\text{new}}) = \text{repr}(p_0) \| \text{repr}(p_1)$$

Encoding new text

Apply learned merges by rank. Earlier learned pairs win when several merges are possible.

Source: Gage, Sennrich et al., GPT-2, and minBPE

A hand-worked merge exposes the algorithm

Corpus: `low low lower`

Round	frequent pair	stream fragment
0	–	`l o w | l o w | l o w e r`
1	`l o`	`lo w | lo w | lo w e r`
2	`lo w`	`low | low | low e r`

What the table hides

Real byte-level training also defines deterministic tie-breaking, special-token policy, and merge boundaries. These are part of the tokenizer artifact.

The “SolidGoldMagikarp” story

1. **Tokenizer data:** a Reddit username appeared repeatedly in counting threads. BPE learned `SolidGoldMagikarp` (with a leading space) as one token.
2. **Pretraining data:** filtering can remove repetitive content from the data used to train the model, while the tokenizer keeps its vocabulary.
3. **The mismatch:** a token frequent enough to be learned by BPE may be rare or absent during model training, leaving the model with a poorly learned representation.

> **Pipeline lesson**
>
> Cleaning model data alone does not clean the tokenizer vocabulary. Compare the datasets and check token exposure.

The exact cause is unconfirmed: the model’s pretraining data is not public.

Source: Rumbelow and Watkins (2023); Land and Bartolo (2024)

Vocabulary size has two opposing effects

Larger vocabulary

fewer tokens per document
lower attention and decode length
better compression for frequent patterns

But also

larger embedding and output matrices
rarer tail tokens
more expensive vocabulary softmax

$$\text{embedding parameters} = Vd \qquad \text{compression} = \frac{\text{UTF-8 bytes}}{\text{tokens}}$$

Weight tying avoids paying Vd twice.

Bits per byte: compare on the same text

Two different questions

Tokenizer compression: how many tokens represent the text?

Model performance: how much probability does the model assign to it?

For the same held-out text, express negative log-likelihood in *bits per byte*:

$$\text{bits / byte} = \frac{\text{total negative log-likelihood (nats)}}{\ln 2 \times \text{UTF-8 bytes}} = \frac{\text{bits}}{\text{token}} \times \frac{\text{tokens}}{\text{byte}}.$$

Example

1,000 bytes, 400 tokens, summed negative log-likelihood 800 nats $\Rightarrow 1.15$ bits / byte.

Compare on identical text and boundary policy. Lower is better; fewer tokens alone do not guarantee a lower model loss.

Checkpoint: tokenizer choice

Think, then compare

A new tokenizer shortens a representative corpus by 20% but adds 300 million embedding parameters. Which metrics on both training and inference would decide whether this is a good trade?

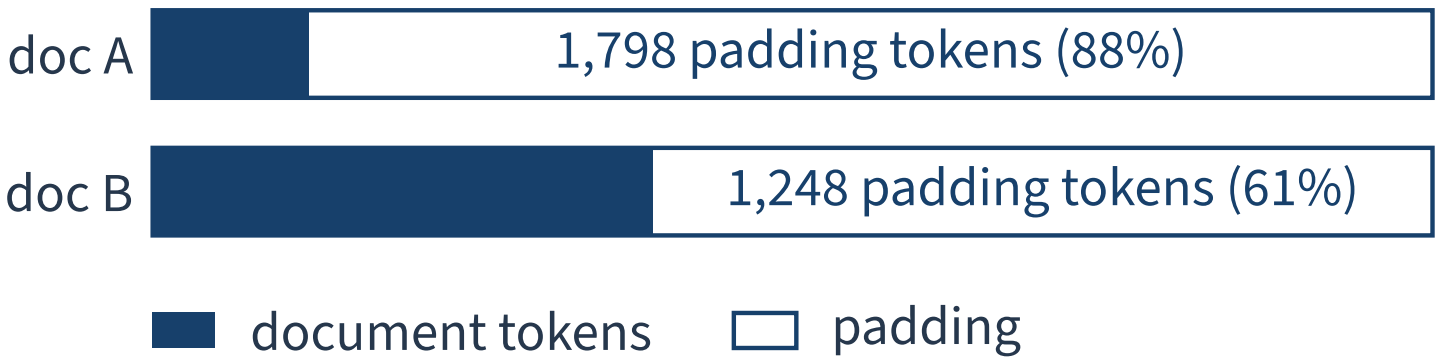
PART 2

Packing documents into examples

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Padding leaves unused positions

Suppose each example is padded to a 2,048-token window:



Packing

Concatenate documents with explicit boundaries and slice fixed-length windows. Padding overhead depends on the model and implementation.

Packing improves utilization but changes attention structure

```
[doc A][EOT][doc B][EOT][doc C]
```

- Every non-masked target can contribute to the loss.
- Windows can cross a document boundary.
- Standard causal attention lets doc B see doc A.

Two valid policies

Accept cross-document context and record it.

Or use block-diagonal causal masks with document offsets.

Invariant

Train and validation splits must happen at the document level before packing.

The end-of-text token has two roles

1. It preserves a recoverable boundary in a contiguous stream.
2. It teaches the model that one document ended and another may begin.

Special-token policy

Reserve its ID before training merges. Prevent ordinary text from being confused with the special marker. Test round-trip behavior explicitly.

PART 3

Storage and correctness

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One contiguous slice creates inputs and targets

Tokens are loaded from a large flat 1D map s. To form an input–target pair of context length T, sample $T+1$ contiguous elements starting at offset i:

$$x = s[i : i+T] \qquad y = s[i+1 : i+T+1]$$

	0	1	2	$\cdots$	$T-1$
x	s_i	s_{i+1}	s_{i+2}	$\cdots$	s_{i+T-1}
y	s_{i+1}	s_{i+2}	s_{i+3}	$\cdots$	s_{i+T}

Fast correctness check

For every batch, $y[:, :-1]$ must equal $x[:, 1:]$ exactly.

A data pipeline needs end-to-end evidence

1. Encode and decode arbitrary Unicode text losslessly.
2. Verify deterministic token IDs and special-token handling.
3. Measure compression on representative languages and domains.
4. Audit document-level split membership before packing.
5. Measure loader throughput; check for GPU input stalls when a trace is available.

Do not optimize a corrupt stream

Decode sampled windows to inspect boilerplate and boundaries; compare document identities across splits to check leakage.

Lab: follow a document into a batch

1. Identify duplicates and keep train and validation documents disjoint.
2. Trace two BPE merges; fit the tokenizer on training text only.
3. Pack each split with EOT delimiters and inspect a boundary.
4. Load a window and verify $y[:, :-1] = x[:, 1:]$.

If time is short

Inspect a provided shard and diagnose a shifted-target error; distinguish inspection from your own implementation.

Checkpoint: pipeline audit

Think, then compare

A training run is fast, but validation loss is implausibly low. List three document- or token-level checks that distinguish a good model from split contamination.

Key takeaways

1. Byte-level BPE guarantees coverage, while merges trade vocabulary size for shorter sequences.
2. Pre-tokenization and special-token rules are part of the learned tokenizer.
3. Packing removes padding waste but requires an explicit document-boundary policy.
4. Binary shards are useful only when their schema, split, and target shift are verified.